



INTRODUCTION

our group has been allotted the Private Jet category and has selected the Gulfstream G650 — a premier ultra-long-range business jet manufactured by Gulfstream Aerospace.

This study presents a complete engine design workflow for the route from Sardar Vallabhbhai Patel International Airport, Ahmedabad (AMD) to Heathrow Airport, London (LHR) — a great-circle distance of 6,900 km. The workflow covers mission analysis, parametric cycle analysis (PCA), and engine performance analysis (EPA) using PERF V5.0, culminating in a charter cost estimate.



REQUEST FOR PROPOSAL

Table 1. Aircraft specifications — Gulfstream G650

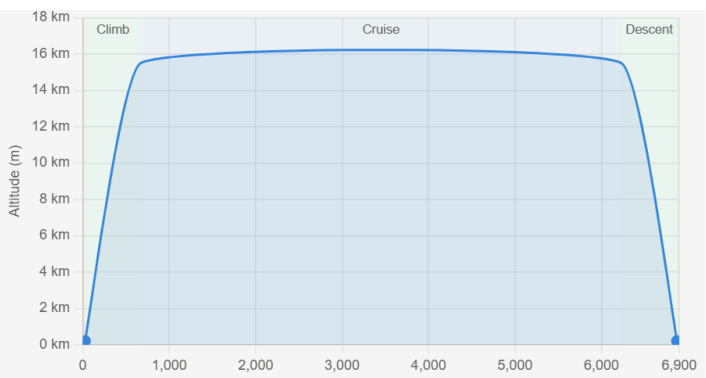
Parameter	Symbol / Unit	Value
Seating capacity	—	13–16
Maximum payload	W_{payload} (N)	28,900
Maximum fuel capacity	$m_{\text{fuel,max}}$ (kg)	20,049
Maximum take-off weight	MTOW (kg)	45,178
Maximum landing weight	W_{land} (kg)	37,875
Wing area	S (m ²)	119.2
Aspect ratio	AR	7.75
Engine type	—	Turbofan ×2 (RR BR725-A1-12)
Max engine inlet diameter	D (m)	1.27
Maximum lift coefficient	$C_{L,\text{max}}$	1.6
Turbine inlet temperature	T_{t4} (K)	1500
Calorific value (Jet-A)	h_{PR} (MJ/kg)	43

Table 2. Journey specifications — AMD to LHR

Parameter	Value
Route	AMD (Ahmedabad) → LHR (London)
Total range	6,900 km
Cruise altitude / Mach	15.5 km / $M_0 = 0.85$
Runway length (AMD)	3,500 m (sea level, $T_{\text{amb}} = 38^\circ\text{C}$)
Single-engine climb gradient	2.4%

MISSION ANALYSIS

Phase	Dist.(km)	Thrust(kN)	Time(s)	Fuel(kg)	W_{start} (kg)	W_{end} (kg)
Taxi (out)	5.0	7.5	500	260	45178	44918
Takeoff	20.0	132.2	53	486	44918	44432
Climb	645.8	33.7	4306	10059	44432	34373
Cruise	5558.4	21.8	22165	3359	34373	31014
Descent	645.8	5.0	3588	1246	31014	29768
Land & Taxi	25.0	7.5	625	325	29768	29443
Total	6900	—	31237	15735	45178	29443



Total fuel burn: 15,735 kg. Fuel reserve = 20049 – 15735 = 4,314 kg (21.5% margin, above the 10% minimum). The climb phase dominates (≈64% of total fuel) due to the OEI-gradient-derived 645.8km climb at near full thrust.

CONCLUSION

A complete engine design workflow was demonstrated for the Gulfstream G650 on the Ahmedabad– London route (6900 km). Mission analysis yielded a fuel burn of 15,735 kg with a 21.5% reserve margin. PCA identified Option B ($\alpha = 4$, $\pi_c = 30$, $\pi_f = 1.7$) as the optimal design, offering a 30% specific thrust margin and $\eta_O = 35.5\%$. EPA using PERF V5.0 confirmed full mission feasibility: takeoff thrust margin +4.8%, OEI climb gradient 4.97% , and a fuel burn of 16,124 kg (+2.5% from Mission analysis). The estimated charter ticket cost is 6.64 lakhs/seat (≈ USD8,000).

REFERENCES

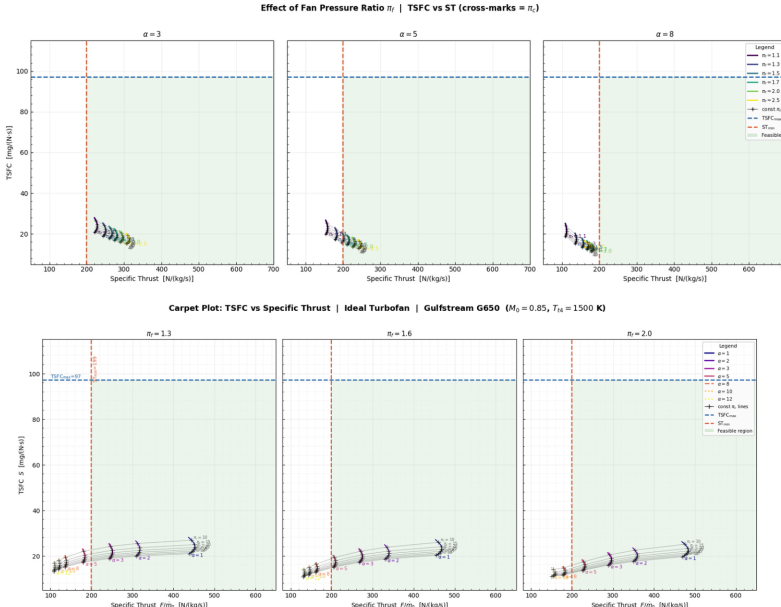
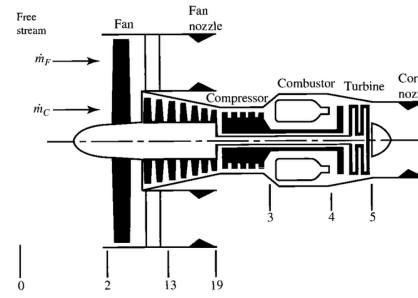
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PARAMETRIC CYCLE ANALYSIS

PCA was performed at the cruise design point ($M_0 = 0.85$, $h = 15,500$ m, $T_0 = 216.65$ K, $T_{t4} = 1,500$ K). The governing ideal turbofan equations were solved over the ranges: $\pi_c \in [10, 40]$, $\pi_f \in [1.1, 2.5]$, $\alpha \in [1, 7]$.

Mission constraints:

- Maximum TSFC: $S_{\text{max}} = 69.4$ mg/(N·s)
- Minimum Specific Thrust: $ST_{\text{min}} = 184$ N/(kg/s)



- Three feasibility conditions:
- (i) $\pi_t > 0$ (turbine not thermodynamically over-expanded),
- (ii) $(V_9/a_0)^2 > 0$ (core nozzle physically feasible),
- (iii) $f > 0$ (fuel is added, not removed).

Table 6. PCA design options — performance at cruise design point

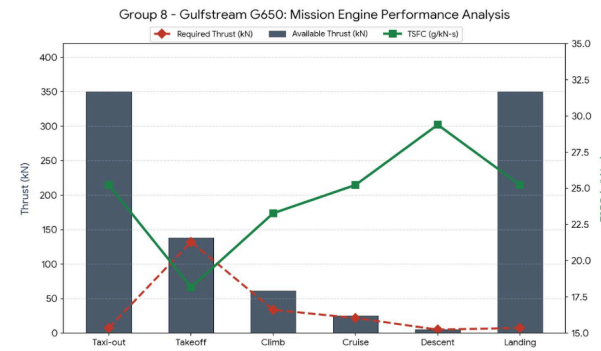
Option	α	π_c	π_f	$F/\dot{m}_0$ [N/(kg/s)]	S [mg/(N·s)]	η_T	η_P	η_O
A — Fuel Economy	5	40	1.8	217	14.2	69.5%	59.2%	41.1%
B — Balanced (rec.)	4	30	1.7	240	16.4	66.9%	53.0%	35.5%
C — Thrust Margin	3	25	1.8	286	17.9	65.2%	49.9%	32.5%
Constraint	—	—	—	> 184	< 69.4	—	—	—

Option B ($\alpha = 4$, $\pi_c = 30$, $\pi_f = 1.7$) is recommended: it provides a 30% ST margin above STmin, uses mature $\pi_c = 30$ technology, and achieves $\eta_O = 35.5\%$. Option A has the best TSFC but only a 9% ST margin, while Option C gives maximum thrust headroom at the cost of efficiency.

ENGINE PERFORMANCE ANALYSIS

Option B parameters were input into PERF V5.0 (Turbofan — 2 Exhausts) with T_{t4} raised to 1,750 K for the non-ideal cycle and a reference mass flow of 50 kg/s per engine.

- Component Efficiencies
- LP Compressor: 88.15%
- HP Compressor: 85.43%
- Fan: 88.15%
- HP Turbine: 90.17%
- LP Turbine: 90.68%



Phase	Condition	Thrust Available	Thrust Required	Margin
Cruise	M 0.85, 15,500 m	12,582 N	10,913 N	+15.3%
Takeoff	M 0.2, Sea Level	69,283 N	66,100 N	+4.8%
Climb	M 0.65, 9,000 m	30,695 N	33,656 N	—
Descent	M 0.7, 7,500 m	2,599 N	2,500 N	+3.9%

Table 8. Final summary of results — EPA-derived TSFC and updated fuel burn

Phase	Dist.(km)	Thrust(kN)	TSFC	Time(s)	Fuel(kg)
Taxi (out)	5.0	7.5	25.23	500	95
Takeoff	20.0	132.2	18.16	53	127
Climb	645.8	33.7	23.28	4306	3377
Cruise	5558.4	21.8	25.23	22165	11880
Descent	645.8	5.0	29.39	3588	527
Land & Taxi	25.0	7.5	25.23	625	118
Total	6900	—	—	31237	16124

OEI ANALYSIS

FAR 25.121 mandates a minimum net climb gradient of 2.4% with one engine inoperative. At sea level, MTOW, and VTO = 77 m/s, the drag with asymmetric thrust and windmilling engine was computed as 31,500 N. With one engine producing 69,283 N:

$$\sin \theta_{\text{OEI}} = (69,283 - 31,500) / 435,878 = 0.0867 \rightarrow \text{Climb gradient} = 4.97\%$$

The achieved OEI gradient of 4.97% exceeds the regulatory requirement of 2.4% by a margin of +107%, confirming the engine design is fully compliant with airworthiness standards under single-engine failure conditions.

TICKET PRICING BREAKDOWN

Fuel cost: 16,124 kg ÷ 0.800 kg/l = 20,155 l at ₹1,50,000/kl → **₹30,23,250**
Fuel constitutes approximately 35% of total charter operating cost. After considering every component cost: Cost per seat (13 passengers): **₹6,64,234 ≈ USD 8,000**
For reference: Commercial business class AMD–LHR ≈ ₹1.8–2.5 lakh; first class ≈ ₹4–5 lakh. The private charter premium reflects exclusive aircraft use and full operational overhead.